

Frontend Specification

Loyalty Programme SaaS

Technical specification document for the web application frontend

1. General Vision

The Loyalty Programme frontend is a web application built with **Next.js** and Server-Side Rendering. It covers two distinct universes that never overlap: the **Administration Dashboard** for tenants who configure their loyalty programme, and the **Member Portal** for end customers who view their points, rewards and wallet. Both universes share the same codebase but have completely separate layouts, routes and authentication logic.

The application is designed **PWA-first** — it must function as an installable mobile app, with a Service Worker managing cache and push notifications. The design is clean, professional and modular. Each tenant can customise primary colours, the logo and the name of their virtual currency — these preferences are loaded from the TenantConfig and applied dynamically via CSS variables at the root layout level.

2. Universe 1 — Tenant Administration Dashboard

This is the interface used daily by the marketing team or loyalty manager of a client company to run their programme. No coding is required — everything is configured through forms, selectors and visual dashboards.

Page 1 — Admin Authentication

This page is the single entry point for all administrators across all tenants. It displays the Yowyob Loyalty platform logo in the centre, a minimal form with an email field and a password field, a primary login button, and a 'Forgot password' link. Below the form, a 'Sign in with' section offers social authentication via Google and Microsoft for tenants that have enabled this option in YowAuth0.

After submission, the system automatically identifies the tenant the administrator belongs to from the claims in the JWT returned by YowAuth0. If the administrator manages multiple tenants (e.g. an external consultant), an intermediate step displays their tenant list as cards with the logo and name of each platform, allowing them to choose which one to work on. This choice is remembered for the session.

The page handles error states cleanly — incorrect credentials, suspended account, inactive tenant — with explicit, non-generic messages. It displays no navigation bar or header until the user is authenticated.

Page 2 — Overview (Home Dashboard)

This is the first page the administrator sees after login. It provides an instant snapshot of the health of their loyalty programme through key indicators organised into clear visual sections.

At the top, a welcome banner displays the tenant name, configured logo, active plan (Free / Pro / Enterprise) and renewal date. An orange alert banner appears when issues require the admin's attention — a member's wallet frozen pending a decision, a campaign expiring in 48 hours, a promo code usage threshold nearly reached.

The central section presents **six metric cards**. (1) Total active members with a sparkline chart showing evolution over the last 30 days and a percentage change versus the previous month. (2) Total points distributed this month, with the same comparison logic. (3) Redemption rate — percentage of distributed

points exchanged for rewards — with a colour gauge (red < 10%, orange 10–30%, green > 30%). (4) Total wallet transaction volume for the month in virtual currency and XAF equivalent. (5) New referred members this month. (6) Active promo codes with their average usage rate.

Below the metrics, two side-by-side panels: the left panel lists the ten most recent important programme events (member upgraded to Gold, promo code exhausted, fraud detected, campaign activated) with a colour-coded icon, a timestamp and a link to the detail. The right panel is a multi-line chart showing day-by-day evolution over the last 30 days of points distributed, points spent and wallet transaction count, with daily / weekly / monthly view toggle.

At the bottom, a 'Quick Actions' block with four wide buttons: Create a rule, Launch a campaign, Add a reward, Generate a promo code — each opens the corresponding form directly without additional navigation.

Page 3 — Member Management

A complete registry of all the tenant's members, displayed in a paginated table with search and advanced filters. The search bar finds members by name, email, phone number or external_id. A Filters button opens a side panel with combinable filters: tier (Bronze / Silver / Gold / Platinum), wallet status (ACTIVE / FROZEN / CLOSED), segment, registration period, points balance range, and recent activity.

Each table row shows: avatar or initials, name and email, tier with coloured badge, formatted points balance, wallet balance in virtual currency, wallet status indicator, registration date and last activity date. Clicking any row opens the member's detailed profile page (its own URL, not a modal). An 'Export CSV' button at the top right exports the currently filtered members.

The member profile page is organised in tabs: **Overview** (all member info, wallet balance and recent transactions, tier progression bar, active rewards); **Transaction History** (all WalletTransactions and PointsTransactions with the same filter capabilities); **Rewards** (all RewardGrants with status); **Referral** (referral link, recruited referees, conversion status, generated rewards); **Admin Actions** (freeze / unfreeze wallet with mandatory reason, manually add points or wallet credit with justification, close account).

Page 4 — Loyalty Rules

The central page of the dashboard — where the administrator configures all programme logic. It has two switchable views: **Templates** and **Custom Rules**.

The Templates view shows a gallery of cards, each representing a ready-to-use template. Each card shows a representative icon, the template name in bold, a one-line description, and a 'Configure' button. Clicking Configure expands the card in place to show only the template-specific parameters — no technical jargon, no visible JSON. A 'Loyalty Card' template asks only how many purchases before the reward and what reward to give. A 'Points Programme' template asks how many points per purchase. A 'Welcome Bonus' template asks how many points to award on sign-up. Once parameters are entered, a 'Create rule' button generates everything in the background and shows confirmation.

The Custom Rules view lists all existing tenant rules in a table: name, status with an on/off toggle, trigger type summarised in one plain sentence, trigger count since creation, creation date, and action buttons (edit, duplicate, archive). A 'New Rule' button opens the Rule Builder.

The **Rule Builder** opens in a dedicated page (not a modal). It is structured in three clearly labelled vertical sections:

WHEN — an event trigger selector with a dropdown listing all supported event types in plain language ('A purchase is completed', 'A member signs up', 'A trip is completed'). Contextual filters appear based on the chosen event type.

IF — conditions panel. An 'Add condition' button adds rows, each with three selectors: metric (cumulative purchase count, points balance, member tier, time of day...), operator (greater than or equal to, equal to...), and value. Conditions are connected by a visible AND operator.

THEN — effects panel. Same logic — 'Add effect' adds rows with a type selector and dynamically appearing parameters.

At the bottom of the Rule Builder: a real-time generated **summary sentence** that updates on every change and describes the rule in plain language — "When a Gold member completes their 10th premium ticket purchase, they receive a free ticket and their counter resets to zero." Plus a collapsible advanced configuration panel exposing priority, validity period and combination policy.

Page 5 — Rewards Catalogue

Administrators create and manage rewards members can obtain. A gallery of cards shows each reward's name, type (distinctive icon), points cost, available stock or 'Unlimited', and active/inactive status.

A creation form slides in from the right and requests: reward name, description shown to members, type among four illustrated radio options (Product or service, Percentage discount, Fixed-amount discount, Wallet credit), points cost, stock with an 'Unlimited' checkbox, validity period of the RewardGrant after attribution, and an optional image uploaded to MinIO. A top filter allows filtering by type, status and sorting by cost or popularity.

Page 6 — Campaigns and Promo Codes

Two tabs: **Automatic Campaigns** and **Promo Codes**.

Automatic Campaigns lists existing campaigns with name, type, validity period with a visual time progress bar, status (Scheduled / Active / Ended) and actions. A 'New campaign' button opens a two-step form: step 1 asks for the name, campaign type (as illustrated cards) and time window (double date-picker); step 2 shows type-specific parameters — for a points multiplier, a slider from 1× to 5× with real-time preview ('A member who earned 100 points per purchase will now earn 200').

Promo Codes lists existing codes with their name, the code itself in monospace on a grey background for easy reading and copying, discount type and value, usage count over total allowed as a colour progress bar (green to 75%, orange to 90%, red beyond), validity period and actions. A 'New code' button opens the creation form.

Page 7 — Tier Configuration

The four tiers are displayed as horizontal cards, from lowest to highest, with a colour gradient (grey for Bronze, silver for Silver, gold for Gold, platinum for Platinum). Each card shows the tier name, icon, points threshold, points multiplier and configured benefits. An 'Edit' button on each card opens an edit form.

Below the tier cards: a 'Maintenance Policy' section to configure required points per period, reference period and grace period. A 'Preview' section shows in real time how current members would be distributed across tiers with the current configuration — a pie chart with percentage and absolute number breakdown.

Page 8 — Referral Programme

Split into two columns. Left: configuration form for the referrer reward, referee reward, conversion condition (which event, minimum amount), maximum conversion delay and anti-fraud policy. Right: real-time programme statistics — active referrals, conversion rate, rewards granted this month, and a top-referrers table.

Page 9 — Wallet and Transactions (Admin View)

A macro view of all tenant wallets. Four top metrics: total volume in circulation (sum of all active balances), top-up volume this month, spending volume this month, number of frozen wallets awaiting decision.

The main table lists all wallets with the same search and filter capabilities as the member page: member ID, name, virtual currency balance and XAF equivalent, status indicator, transaction count this month, and quick actions (view detail, freeze, manual adjustment).

A **Suspicious Transactions** tab shows transactions flagged by the fraud-detection system with reason, member involved, and Validate / Block buttons.

A **Reconciliation** tab runs an on-demand verification and displays whether each wallet balance matches the sum of its transactions. Discrepancies are highlighted in red with details.

Page 10 — Analytics

A comprehensive analytical dashboard organised in clearly titled sections.

Loyalty — stacked area chart of member count per tier over time, monthly retention rate, average member lifetime before churn, most triggered rules as a ranking.

Rewards — most redeemed rewards as a horizontal ranking, total points cost of redemptions per month, average time between reward grant and use, expiry rate of unused rewards.

Wallet — cumulative balances in circulation curve, credit breakdown by source (MTN / Orange / Stripe / Loyalty / Cashback), average transaction value, wallet usage frequency per active member.

Referral — referral funnel from link generation to conversion and reward, with conversion rates at each step.

A period selector (7 days / 30 days / 90 days / 12 months / custom) applies to all charts. An 'Export PDF report' button generates a formatted report with all visible charts.

Page 11 — Tenant Configuration

Global tenant settings page, accessed via a left-side vertical secondary navigation.

Identity — modify tenant name, upload logo, choose primary dashboard colours via a colour picker, enter legal and contact information.

Wallet — configure the virtual currency (name, symbol, conversion rate), enable/disable permitted operations via toggles, set all WalletPolicy limits via numeric fields with clearly indicated units.

Notifications — configure message templates for each event type (credit, debit, reward, tier downgrade...) with dynamic variables insertable via clickable chips ({member_name}, {amount}, {reward_name}...). A 'Send test' button previews the notification on the admin's own account.

Integrations — tenant API keys with ability to generate new ones and revoke existing ones, list of configured webhooks with URL and listened events, journal of recent API calls with status for debugging.

Security — configure OTP threshold for wallet transactions, session policies, and allowed IP addresses for API calls.

3. Universe 2 — Member Portal

The interface used by the end customer of a client platform to track their loyalty. It can be integrated as a page within the client platform via an embedded widget, or accessed directly as a standalone PWA. It inherits the tenant's visual customisation — colours, logo, virtual currency name.

Page 12 — Member Dashboard

The member's home page. At the top, a visual hero card shows their first name, avatar, current tier with corresponding icon, and a progress bar toward the next tier with the exact number of missing points. The card colour changes by tier — grey for Bronze, silver for Silver, gold for Gold, platinum for Platinum.

Below, two large side-by-side cards: points balance (exact formatted number and equivalent in available rewards) and wallet balance (virtual currency with a direct 'Top up' button).

A 'My available rewards' section shows the first three active unused rewards with attractive visual presentation — image, name, expiry date highlighted in orange if approaching, and a 'Use' button. A 'See all my rewards' link leads to the dedicated page.

A 'Recent activity' section chronologically lists the ten most recent points and wallet movements with a colour-coded icon (green arrow for credit, red for debit), action description, amount and date.

Page 13 — My Wallet

Dedicated wallet management page. At the top, a large card shows the current balance prominently, the virtual currency name, wallet status and an 'Equivalent to X XAF' indication if the conversion rate differs from 1.

Three main action buttons: **Top up**, **Pay** and **Withdraw** (if enabled by the tenant). Top up opens a panel with available top-up options — each provider shown with its logo (MTN, Orange, Stripe), an amount field and limits shown as subtext. Withdraw opens a form with destination account, amount and a summary before confirmation.

Below the buttons, a complete wallet transaction history in a table filterable by type and period. Each transaction can be expanded to show all details — reference, precise timestamp, status, and link to the associated order if available.

Page 14 — My Points and Rewards

Two tabs. **My Points** shows available balance, total earned since registration, total spent, and a visual timeline of the most recent point awards — when, for what action, how many. If points are expiring soon, an orange alert signals them with the remaining time.

The **Catalogue** tab shows all rewards available in the tenant's catalogue, filterable by type and sortable by cost ascending or descending. Each reward card shows an image, name, short description, prominently displayed points cost and remaining stock if defined. The 'Redeem' button triggers a confirmation modal showing the cost, remaining balance after redemption, and asking for confirmation.

Page 15 — My Tier

Details the member's progression in the tier programme. At the top: current tier, date achieved, active multiplier. Below, a horizontal visualisation of all tiers with each tier's benefits in a checklist — already-acquired benefits are checked in green, next-tier benefits are greyed out with a padlock.

A 'How to progress' section explains in simple terms how many more points are needed for the next tier and suggests concrete actions to get there quickly.

If the tenant has configured a maintenance policy, a 'Maintaining my tier' section shows points accumulated in the current period, the threshold to reach, and the time remaining before the period ends — with a colour progress bar.

Page 16 — Referral

Presents the tenant's referral programme engagingly. At the top, a Hero block explains the benefits — what the member earns and what their referee earns — with prominently displayed amounts.

Below, the member's referral link in a highlighted box with a 'Copy' button and direct share buttons for WhatsApp, SMS and other available channels. A downloadable QR code is also provided.

The 'My referees' section lists all referred contacts with their partial name or email (for privacy), their status (Registered / Awaiting conversion / Converted), and the reward already received or pending. A counter at the top of this section proudly displays the total number of converted referees.

4. Cross-Cutting Technical Notes

Each page uses Next.js's nested layout system — a root layout that loads the TenantConfig and applies CSS variables, an admin layout with sidebar navigation, and a member layout with a bottom mobile navigation bar.

Data is loaded server-side via Next.js **Server Components** for static or infrequently updated data (configuration, catalogue), and via **Client Components with SWR** for real-time data (balance, notifications). **WebSockets or Server-Sent Events** handle real-time balance and notification updates without requiring the member to refresh the page.